From Point Clouds to Prop Roots: Evaluating Automated vs QSM TLS Workflows Across Tidal Forested Wetlands

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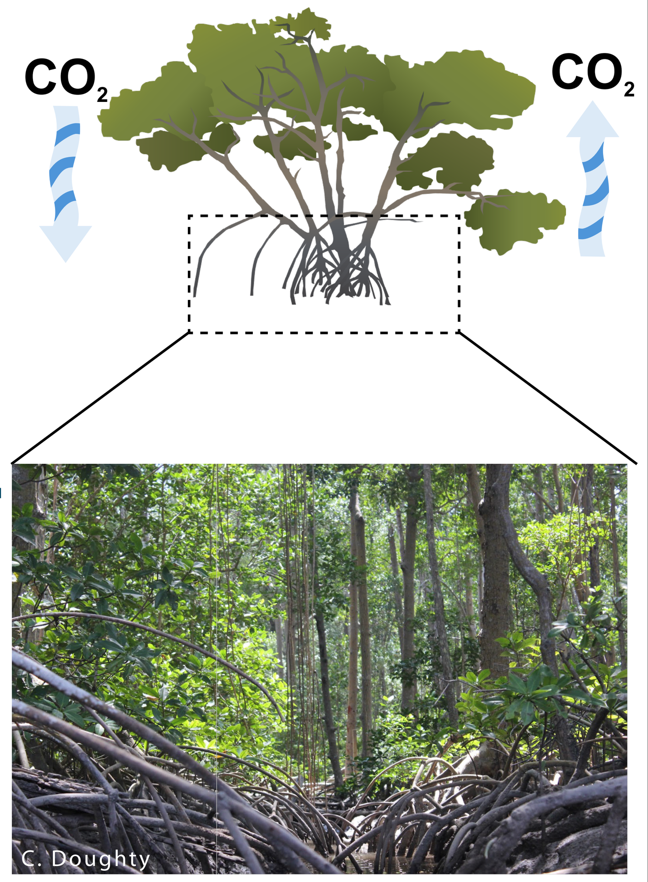
Context

- Mangrove forests are critical blue carbon ecosystems
- Complex structure (i.e., prop roots, dense branching) complicates structural estimation using traditional methods
- Accurate structure is needed for carbon accounting and greenhouse gas flux scaling



Motivation

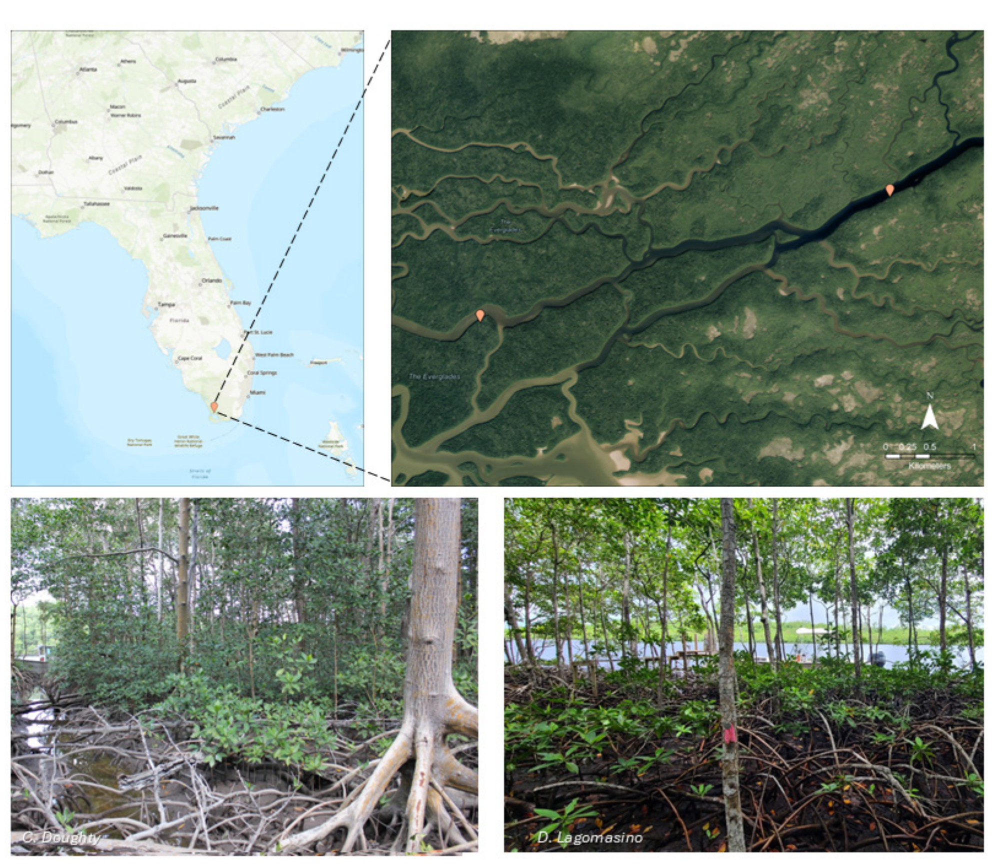
- Terrestrial laser scanning (TLS) captures complex structure and biomass non-destructively
- Traditional segmentation is labor-intensive and requires manual, individual-tree extraction
- Standard methods may miss near-ground root structure

Research Questions & Goals

- Can automated plot-level segmentation produce consistent estimates of woody volume and surface area compared to individual-tree QSM?
- How consistent are estimates across structural components (prop roots, branches, and main stem)?
- How do TLS-derived aboveground biomass from both approaches compare to available allometric equations?

Study Area

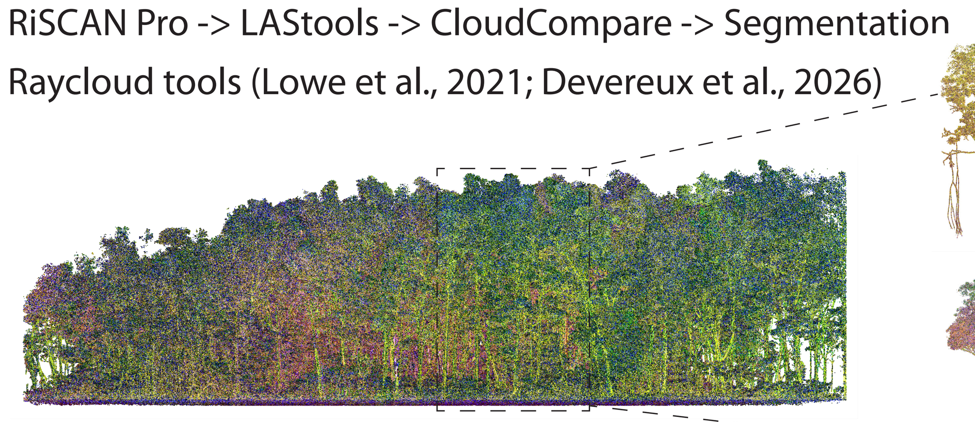
- Shark River Slough, Everglades National Park, Florida
- Subtropical mangrove forests across environmental gradients



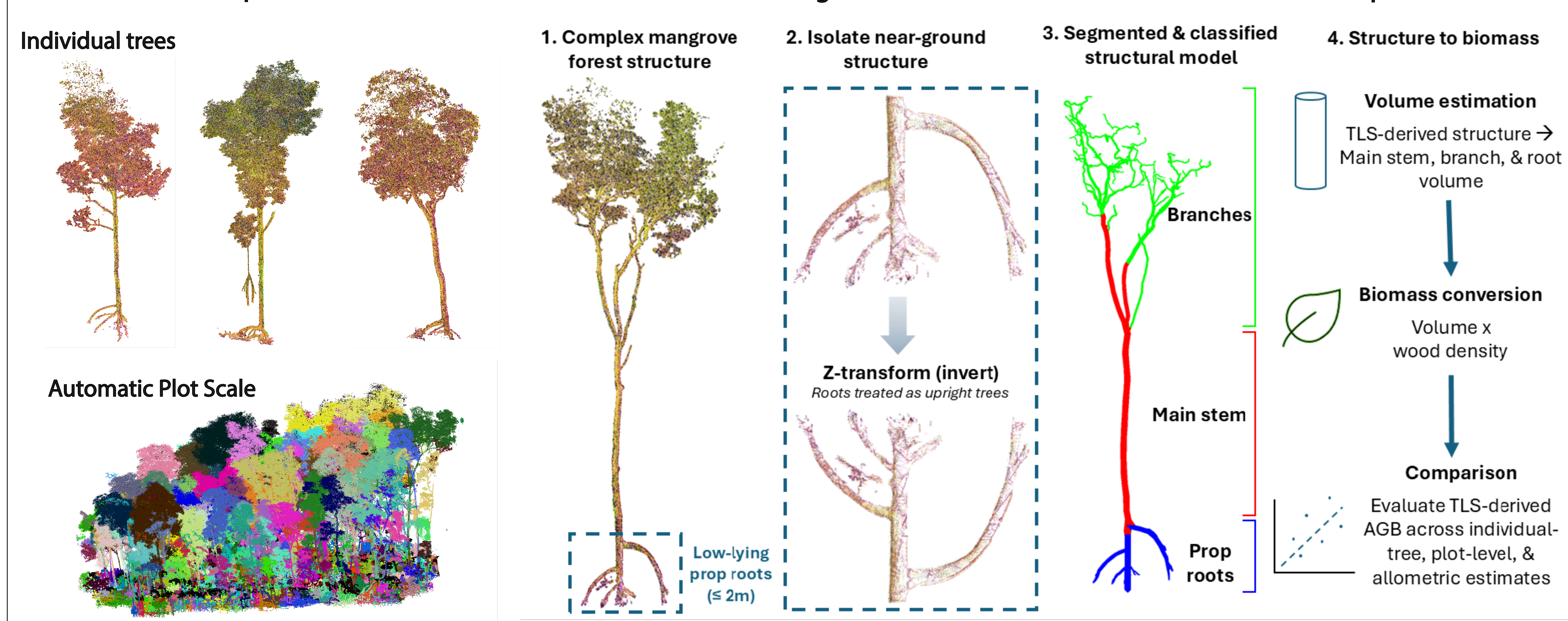
- Dominate species at these sites:
  - Rhizophora mangle
  - Avicennia germinans
  - Laguncularia racemosa

TLS Data Acquisition

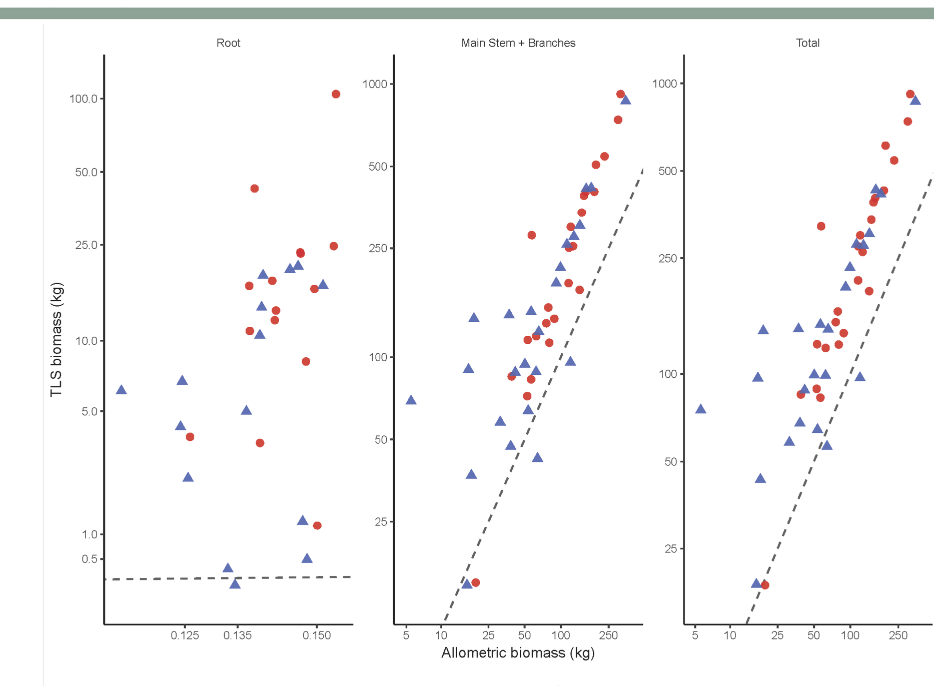
- RIEGL VZ-400i TLS
- 4 scans per plot (~10 m radius)
- GNSS georeferenced (~10 cm accuracy)
- RiSCAN Pro -> LAStools -> CloudCompare -> Segmentation
- Raycloud tools (Lowe et al., 2021; Devereux et al., 2026)



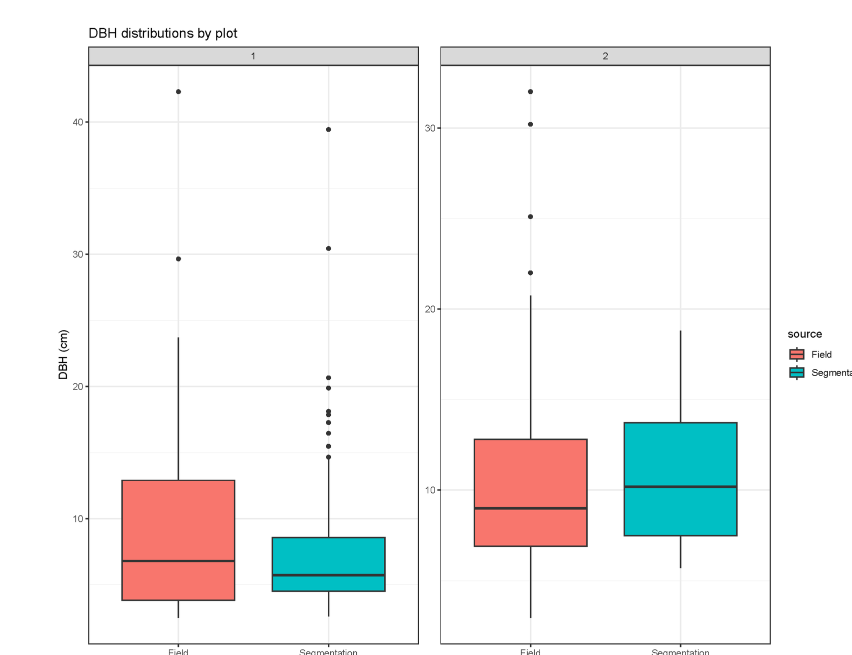
From point clouds to structure



Results

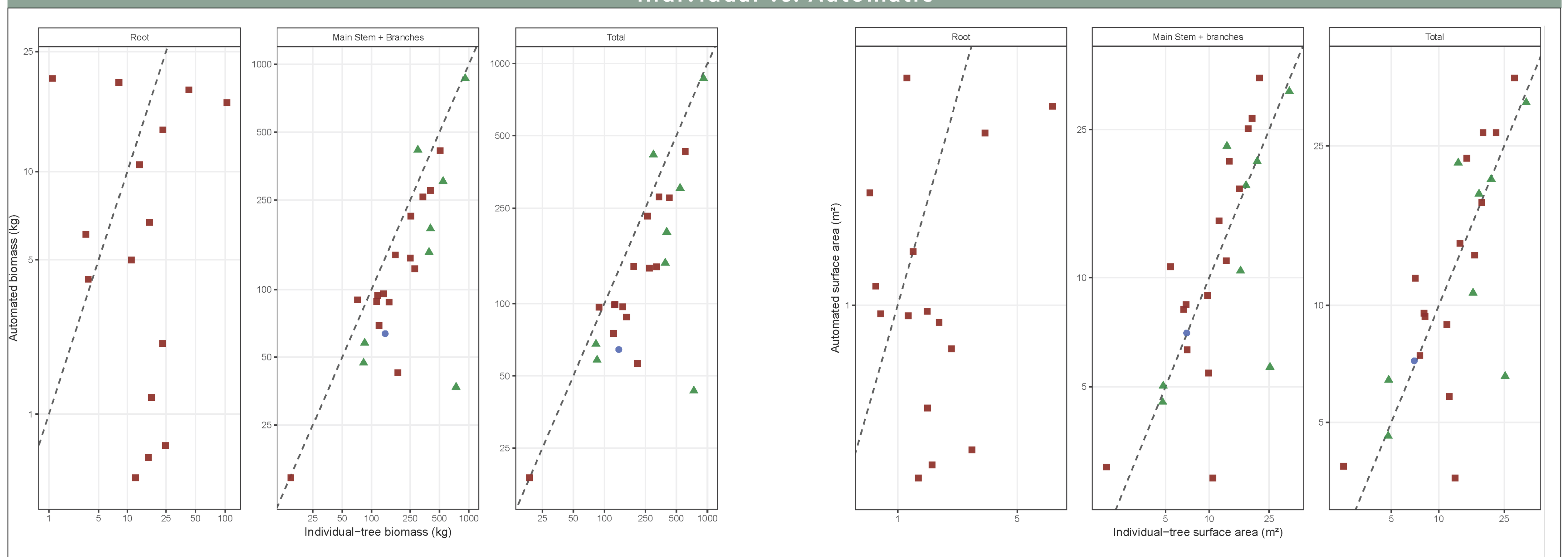


There is some inconsistency between individual and plot-level methods but both methods still capture more biomass than allometry, particularly for roots



TLS-derived DBH distributions are consistent with field measurements across plots, supporting the use of TLS for plot-level structural characterization

Individual vs. Automatic



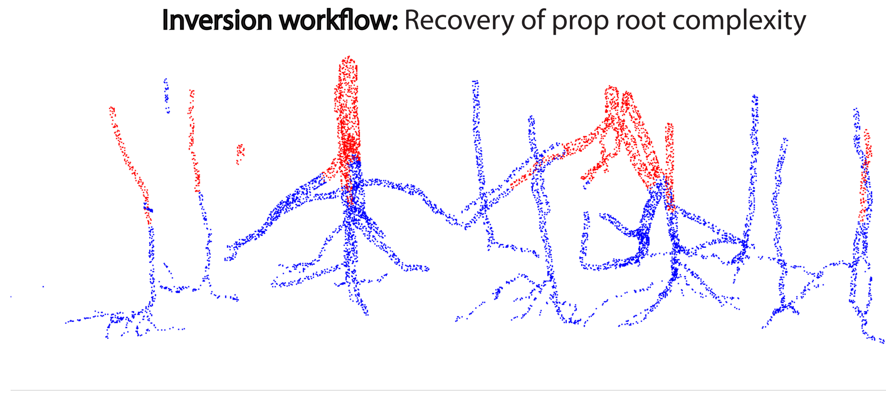
Metric	Component	Units	n	Mean (Indiv)	Mean (Auto)	RMSE	Bias	R^2
Biomass	Root	kg	15	21.51	8.45	26.67	-13.06	0.088
	Main Stem + Branches	kg	24	279.86	177.86	180.72	-102	0.539
	Total	kg	24	293.3	184.98	184.3	-108.32	0.548
Surface area	Root	m^2	15	2.01	1.12	1.79	-0.9	0.107
	Main Stem + branches	m^2	24	13.12	13.68	6.31	0.56	0.53
	Total	m^2	24	14.38	14.58	5.99	0.19	0.588
DBH	Tree-level	cm	24	19.49	15.07	7.32	-4.43	0.359
Height	Tree-level	m	24	14.21	15.57	2.67	1.36	0.653

Key take aways

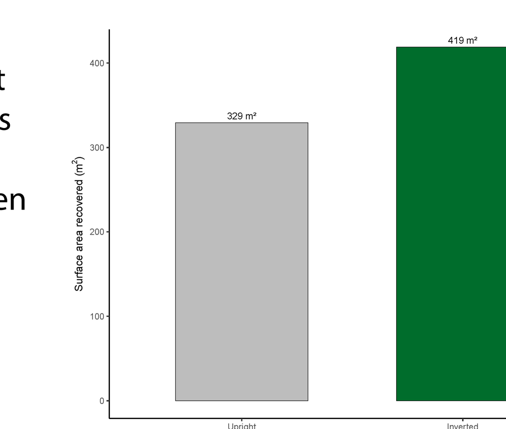
Inversion-based approaches adapted from Asmath et al. (2025) better captures complex root architecture often missed in standard workflows

Standard upright workflow: Minimal prop root complexity

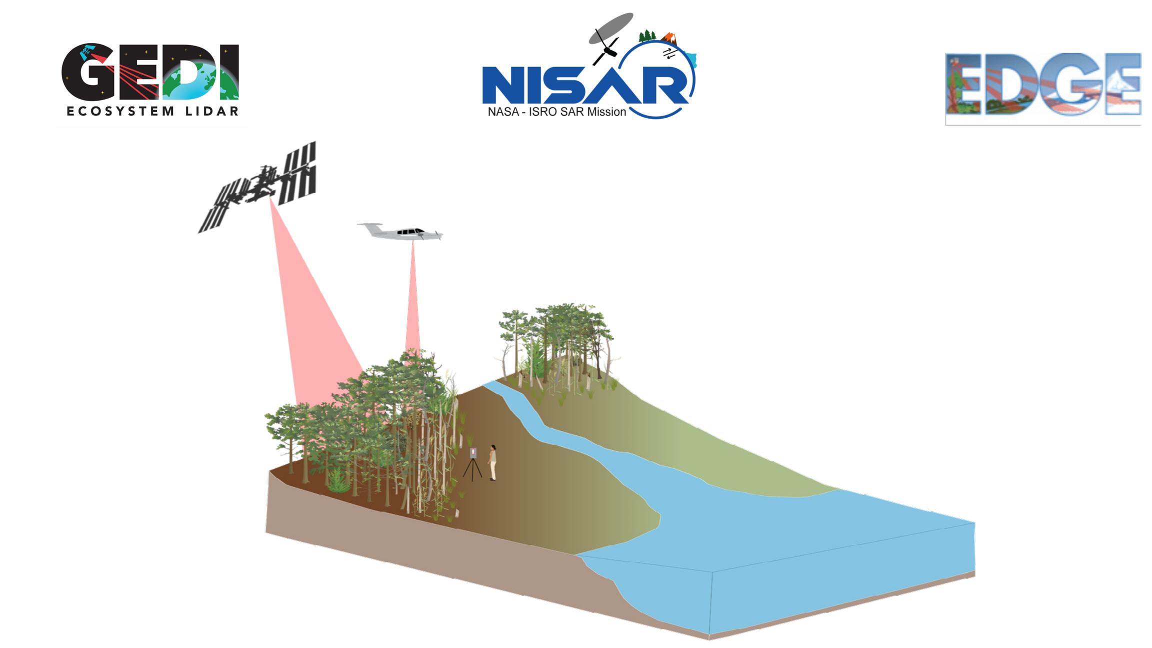
Inversion workflow: Recovery of prop root complexity



- Inversion-based processing recovers substantially more near-ground prop-root structure than standard upright workflows
- Important as near-ground structure is often underrepresented
- Major implications for scaling:
  - Greenhouse gas emissions
  - Carbon accounting



Broader Impacts



- TLS-based workflows capture complex mangrove structure non-destructively
- Improve plot-level biomass estimates beyond traditional allometry
- Enable scalable calibration of airborne and satellite observations without manual tree extraction

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